

Layer-wise Learning Rates & Domain Adaptation

How much should each layer learn when moving to a new problem?

CSE471 — DEEP LEARNING FOR COMPUTER VISION
WEEK 7 | LECTURE 13 | UNIT III



Where We Left Off



Question:

Do all trainable layers need the same learning rate?

Learning Objectives



- ✓ Explain the idea of a learning rate in simple terms.
- ✓ Explain why different layers may need different learning rates.
- ✓ Understand layer-wise learning rates.
- ✓ Grasp the relationship between freezing, unfreezing, and learning rates.
- ✓ Explain what a domain is in machine learning.
- ✓ Explain domain shift.
- ✓ Explain domain adaptation.
- ✓ Identify practical situations where domain adaptation may be useful.

What Is a Learning Rate?

The learning rate controls **how much** a model's parameters change during each training update.



Small Steps

Cautious Updates

The model learns slowly but steadily, preserving existing knowledge without wildly overshooting.



Large Steps

Aggressive Updates

The model changes rapidly. It can learn fast, but risks destroying existing knowledge or missing the target.

Why Does LR Matter for Transfer Learning?

A pretrained model already contains highly useful knowledge.

Pretrained Knowledge

✓ Small Controlled Adjustment

Using a small learning rate safely adapts the model while preserving the useful features it already knows.

⚠ Large Destructive Update

Using a massive learning rate can violently overwrite and wipe out the useful pretrained representations entirely.

Should Every Layer Learn Equally?



Question: Should Layer 1 and Layer 4 receive exactly the same learning rate?

- 💡 **Intuition:** Earlier layers often contain more general representations (edges, simple textures).
 - 💡 **Intuition:** Later layers are often more task-specific (complex object parts).
 - ➔ Therefore, different layers may benefit from different amounts of adaptation!
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The Core Idea: Layer-wise Learning Rates

Assign different learning rates to different parts of a neural network during training.

Early Layers: Very Small LR



Middle Layers: Small LR



Later Layers: Larger LR



New Classifier: Largest LR

Why Different Learning Rates?



Early Layers

Often learn more general visual patterns (edges, shapes, colors). They usually require very little adjustment, so they need **less change**.



Later Layers

Often more task-specific (recognizing specific object parts like tires or eyes). Depending on the new task, they **may need more adaptation**.



New Classifier

Usually initialized from scratch for the completely new task. It **needs substantial learning** to become useful.

Simple Numerical Example

Here is how you might configure a network in code (PyTorch/TensorFlow):

Network Part	Example Learning Rate
Early layers	0.00001 (1e-5)
Middle layers	0.00005 (5e-5)
Later layers	0.0001 (1e-4)
New classifier	0.001 (1e-3)



These are purely **illustrative values** — not universal settings. Exact values depend heavily on the specific model, dataset, optimizer, and experiment.

Connection to Freezing

How does this connect to our previous lecture on freezing layers?



Frozen Layer

Learning Rate: **0**



Trainable Early Layer

Learning Rate: **Small**



Trainable Later Layer

Learning Rate: **Larger**



New Classifier

Learning Rate: **Often
Largest**



Visualizing the Whole Strategy



Summary: Preserve useful knowledge where possible, adapt more where necessary.

Problem Scenario: When Things Go Wrong

Model Pretrained On:

Everyday Photographs



Now Want to Use For:

Satellite Imagery



Question: Will the pretrained model automatically work perfectly?

Answer: Not necessarily. The new data looks very different from the original data.



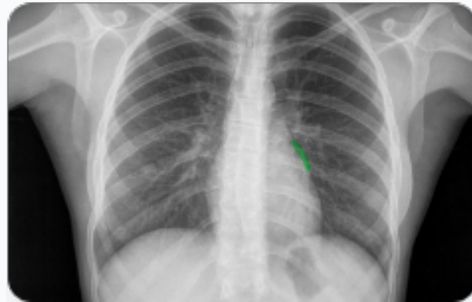
What Is a Domain?

A domain is the type or distribution of data from which a machine-learning problem is drawn.

Domain A
Everyday Photos



Domain B
Medical X-rays



Domain C
Satellite Images

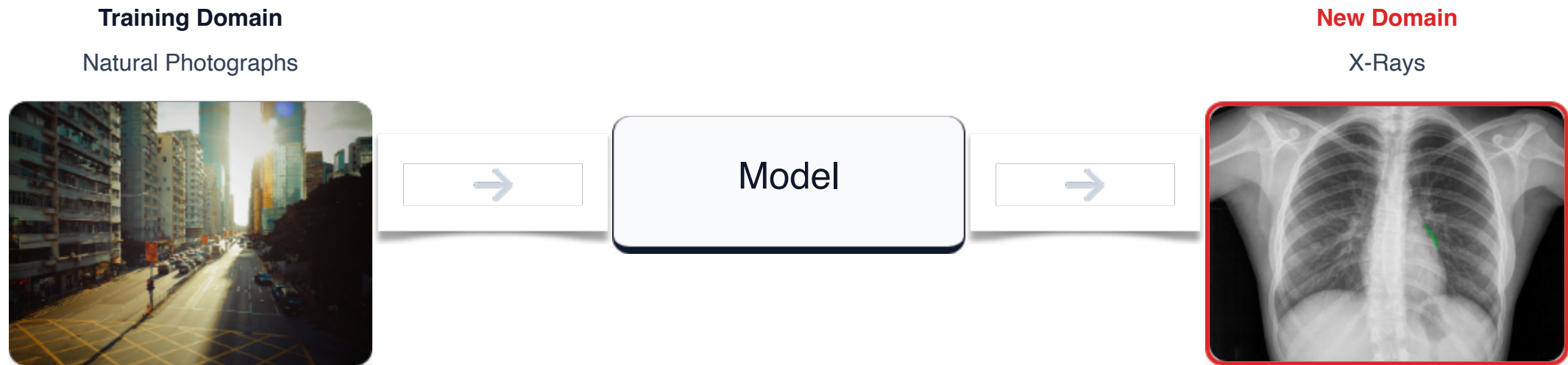


Domain D
Microscope Images



What Is Domain Shift?

Domain shift occurs when the data distribution in the new environment differs from the distribution seen during training.



Highlight: The underlying model architecture is the same, but the input data has changed significantly.

Real-World Example: Day vs Night

Model Trained On:

☀️ Daytime Road Images



Model Tested On:

🌙 Nighttime Road Images



The Task is exactly the same (e.g., Object Detection for cars), but the visual conditions have changed: **Lighting, Shadows, Visibility, Image Characteristics.**

More Examples of Domain Shift



Camera Change

Camera A ➡ Camera B
(Different lenses, sensors, color profiles)



Lighting Change

Sunny ➡ Overcast ➡ Night
(Shadows, contrast, reflections)



Geography

One Country ➡ Another Country
(Different road signs, architecture styles)



Environment

Clear Weather ➡ Rain / Fog / Snow
(Obscured vision, noise, atmospheric scattering)

In all cases, the underlying task remains the same while the input distribution changes.

Why Domain Adaptation?

What can we do if our model works well on the source domain but poorly on the target domain?

Domain Adaptation

Domain adaptation aims to make a model trained on a source domain work better on a different but related target domain.



Source Domain vs Target Domain

Source Domain

Where the model has learned from.

Example

Large collection of daytime road images.

Target Domain

Where we want to use the model.

Example

Nighttime road images.

Goal: Reduce the gap between source and target domains.

Domain Adaptation vs Transfer Learning



Transfer Learning

Broad Idea:

Reuse knowledge from one task or domain to help solve another.

(e.g., ImageNet classification → Dog/Cat classification)

Domain Adaptation

Specific Situation:

Adapt a model to a different target domain, typically while keeping the same task.

(e.g., Daytime car detection → Nighttime car detection)

Domain adaptation can be considered a **specific setting within the broader idea** of transferring knowledge.

How Can We Adapt?

1. Fine-tuning

Use target-domain data to adjust the pretrained model directly.

2. Data adaptation

Make source and target data more similar where possible (e.g., color matching).

3. Feature adaptation

Encourage the network to learn useful representations that work across both domains.

4. Domain-specific training

Use available target-domain data heavily to improve performance directly.

Layer-wise Learning Rates + Domain Adaptation

Scenario:



Pretrained model



Source: Natural photographs



Target: Specialized images

Strategy:

Early layers: Small LR / possibly frozen
(Keep general edge/texture detection)

Later layers: Larger LR
(Adapt to new specialized shapes)

New classifier: Higher LR
(Learn the new specialized classes)

Layer-wise learning rates provide a controlled way to adapt different parts of the model when moving toward the target domain.

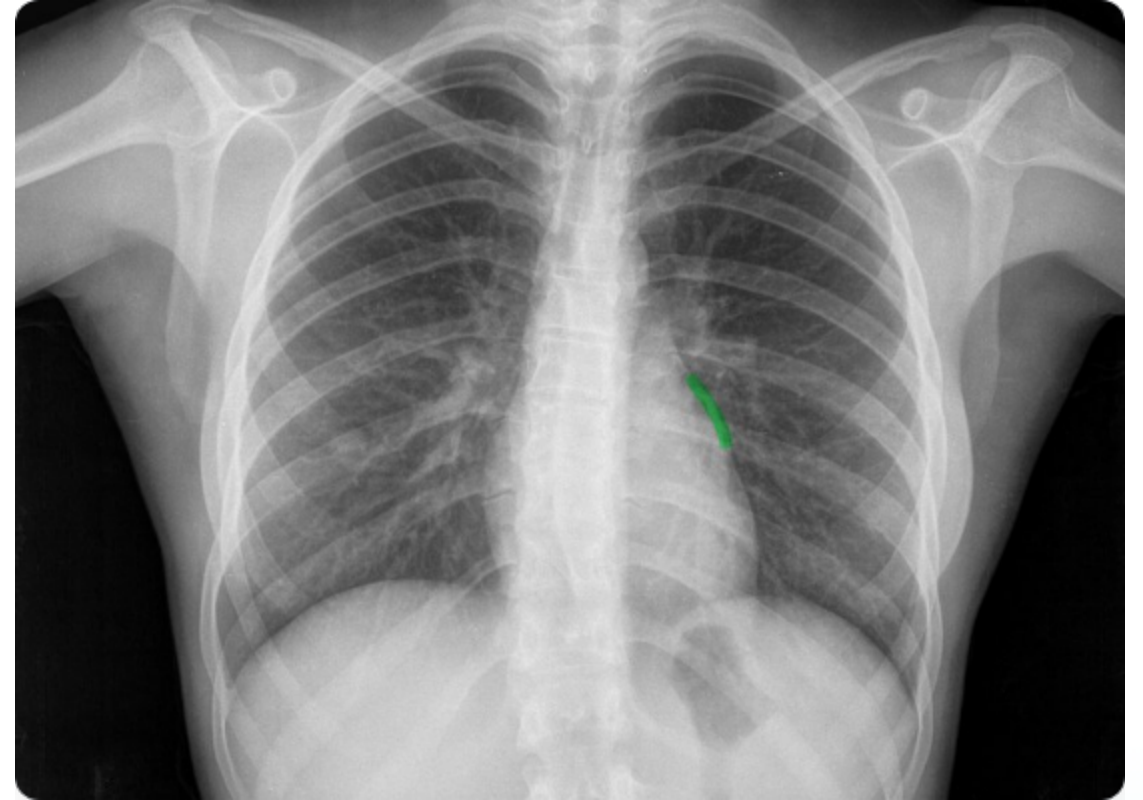
Case Study: Medical Imaging

Scenario: Model pretrained on general photographs.
Target domain is Medical X-rays.

Ask Yourself:

Which parts of the network would you expect to preserve, and which might need more adaptation?

- ▶ General visual representations (edges) may still be somewhat useful.
- ▶ Domain-specific patterns require heavy adaptation.
- ▶ Target data size dictates learning rate limits.
- ▶ Validation is essential!



Case Study: Autonomous Driving



Trained: Clear daytime roads

Deployed: Rain + night + different camera

What has changed?

- ☁️ Lighting & Weather
- 📷 Camera hardware
- 👁️ Image appearance

Discussion:

Would simply freezing the entire backbone always be enough here?

A Simple Strategy (Workflow)



When approaching a new transfer learning problem with a domain shift:

- 1 Start with a pretrained model.
- 2 Understand your source and target domains.
- 3 Start simple: Try feature extraction / conservative fine-tuning first.
- 4 Use smaller learning rates for pretrained layers.
- 5 Use a larger learning rate for the new classifier when appropriate.
- 6 Evaluate heavily on target-domain validation data.
- 7 Gradually unfreeze and adapt more layers if necessary.



Final Recap + Exit Question

Key Terms:

Learning Rate: Controls update size.

Layer-wise LR: Different layers can use different learning rates.

Domain: The data environment/distribution.

Domain Shift: Source and target data differ.

Domain Adaptation: Adapting a model to work better on the target.

Big Idea

Preserve what is useful. Adapt what needs to change.

Exit Question:

A CNN was trained on daytime road images but must work on nighttime images. Would you freeze everything, use the same learning rate everywhere, or gradually adapt different parts of the model? Why?

➡ **Next lecture: Dataset considerations — small data, class imbalance, and dataset splitting**

Image Sources



[https://images.unsplash.com/photo-1761877282712-ae09984c8cc1?](https://images.unsplash.com/photo-1761877282712-ae09984c8cc1?fm=jpg&q=60&w=3000&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D)

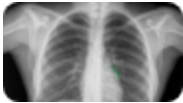
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Source: www.magnific.com



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